

Alessandro Caula

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Profile

Versatile Software Developer and Bioinformatician with expertise in C#, Python, and JavaScript, and a passion for creating efficient, data-driven solutions for scientific applications. Skilled in software engineering, algorithm design, data processing, and visualization, I build robust user-friendly applications. Committed to continuous improvement, my adaptability, creativity, and clear communication have helped me develop tools that enhance research and support users.

Technical Skills

Programming Languages: C#, Python, JavaScript, TypeScript, Bash, HTML, CSS

Frameworks & Libraries: .NET, React, Node.js, Tailwind CSS, Biopython, SciPy

Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, Keras, Nextflow

Visualization: Matplotlib, Seaborn, Plotly, DevExpress, LightningChart

OS & Tools: Windows, Linux, macOS, Git, VS Code, Google Colab, AWS, MongoDB, Docker

Experience

Bioinformatics Software Developer, 3Brain AG – Genova, ITALY

May 2022 – Sept 2025

Designed and developed high-performance scientific software for real-time electrophysiological data acquisition, analysis and visualization in *BrainWave*, a C#/.NET application used by neuroscience researchers. Key contributions:

- Developed and optimized algorithms for signal processing, frequency analysis (3× faster) and trajectory detection.
- Efficiently processed large datasets (50–80 GB/recording) through scalable pipelines, ensuring high performance.
- Engineered modular, object-oriented components, ensuring scalability and maintainability.
- Built interactive and intuitive GUIs and charts for real-time data exploration and visualization.
- Implemented and integrated statistical and machine learning algorithms into core pipelines.
- Worked in a large, fast-moving codebase, collaborating across teams to deliver maintainable, robust software.
- Improved performance and robustness through rigorous debugging and architectural refactoring.

Data Scientist Intern, University of Lisbon – Lisbon, PORTUGAL

Apr 2021 – Feb 2022

Developed and implemented supervised and unsupervised Machine Learning pipelines for omics data analysis using Python, aiming to identify molecular subtypes of Alzheimer's disease ([*GitHub*] [*Paper*]).

- Preprocessed and cleaned data, performed feature selection and modeling, and evaluated results with key metrics.
- Created visualizations and summaries of model outputs, clearly communicating biological insights to researchers.
- Applied a range of ML methods including Non-negative Matrix Factorization, Support Vector Machines, Naïve Bayes, k-Nearest Neighbor, Decision Trees, Logistic Regression, Random Forests, and XGBoost.

Laboratory technician, IHMA Sarl – Monthey, SWITZERLAND

Feb 2019 – Jul 2019

In vitro studies of new and existing antimicrobials for Phase II, III, and IV programs.

- Performed sample processing, organism identification (MALDI-TOF), and antimicrobial susceptibility testing.
- Prepared and validated testing panels, maintained quality control, and supported data recording.

Education

University of Bologna, M.Sc. in Bioinformatics

Oct 2019 – Feb 2022

Gained a strong foundation in computational biology, algorithms implementation, data science, and machine learning.

- **Key coursework:** Algorithms and Data Structures, Python and Bash programming, Basic and advanced Machine Learning, Systems and in Silico Biology, Big Data Processing Infrastructures, DNA/RNA dynamics

University of Turin, B.Sc. in Biology

Sept 2015 – Oct 2018

- **Key coursework:** Cellular and Molecular Biology, Biochemistry, Human and Plant Physiology, Microbiology, Chemistry, Public Health, Statistics and Scientific Analysis.

Languages

Italian - Native

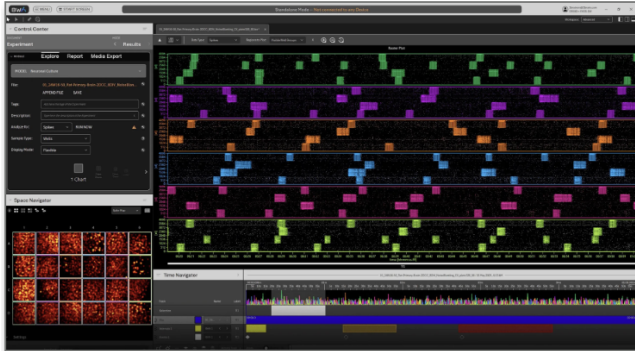
English - Fluent (professional proficiency)

French - Beginner

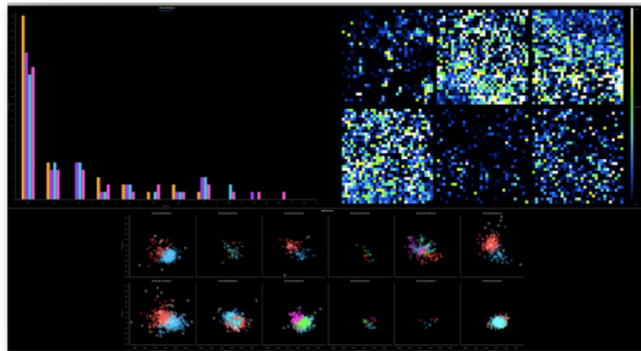
Project Highlights

BrainWave - 3Brain AG

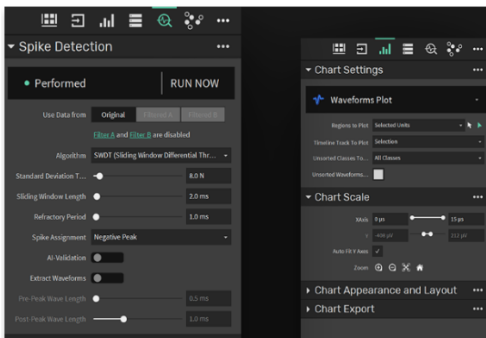
High-performance C#/.NET software for real-time electrophysiology data acquisition, analysis, and visualization.



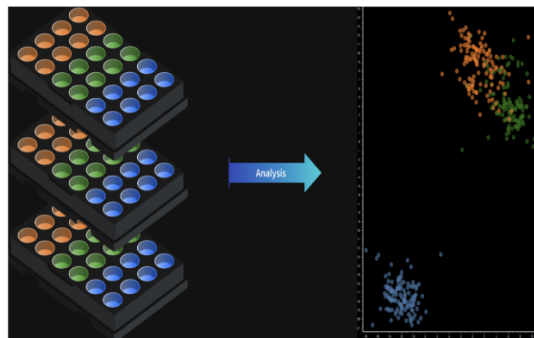
Modular, high-performance software with interactive GUI components.



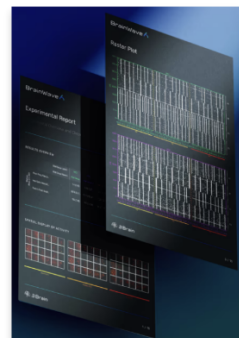
Interactive and dynamic visualizations of electrophysiological analysis.



Fully customizable and intuitive graphical interface.



ML and clustering on multi-channel recordings.



Automated Report.

Bioinformatic pipeline - Protein Loop Reconstruction

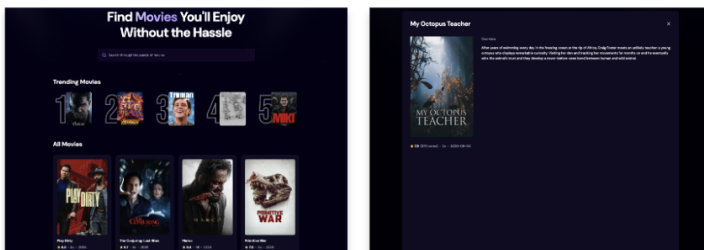
Python pipeline for automatic identification and modeling of missing internal loops in protein structures:

- Generated multiple loop conformations with the CCD algorithm, and evaluated for steric clashes, Ramachandran outliers, and anchor RMSD.
- Designed modular, extendable pipeline for structural bioinformatics applications.

Web Development

Practiced front-end and full-stack web development using **React.js** and modern libraries, applying best practices for modular and maintainable code.

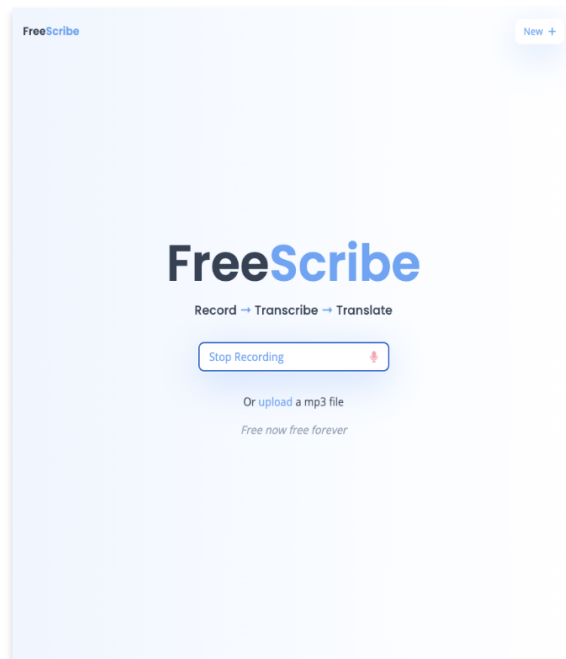
- Worked with APIs, backend communication, and real-time processing (including Web Workers for AI tasks).
- Strengthened skills in modular frontend architecture and user-friendly interface design.



React app for browsing and discovering movies via external API.



Modern landing page built with React, Tailwind, and GSAP.



React app for audio-to-text translation using OpenAI.